MARRI LAXMAN REDDY
INSTITUTE OF TECHNOLOGY AND MANAGEMENT
(AN AUTONOMOUS INSTITUTION)
(Approved by AICTE, New Delhi & Affiliated to JNTUH, Hyderabad)
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Abstract Proforma

Field based Project/ Industry Oriented Mini Project/Major Project

Academic Year: 2025-2026			**Regulation:MLRS-R22**	
Branch :CSE(AIML)	**Year/Sem : IV-I**		**Section: C**	**Project Batch No.: C22**
Registration Details	**S. No**	**Name of the student**	**Roll Number**	**Signature**
	1	Ranesh Gupta	227Y1A66G2	
	2	Sujal Gupta	227Y1A66I1	
	3	Sangana Tanuja Reddy	227Y1A66I2	
Name of the Guide & Designation		Mr.CH.V.Krishna Mohan (Assistant Professor Dept-CSE_AIML)		
Area (Domain) of the Project		Deep Learning, Financial Forecasting (FinTech).		
Title of the Project (Tentative)		StableStock - AI-Powered Stock Prediction & Trading Assistant		
Type of the Project		Application/ Product/ Research/ Interdisciplinary/ Other		

Abstract :
This project introduces a hybrid deep learning framework to tackle the challenge of stock price prediction in volatile markets. The model combines a BiLSTM to capture short-term sequential dependencies and a Transformer to apply global attention for a more robust understanding of long-range patterns. This approach aims to overcome the limitations of traditional and single-method models, providing more accurate and stable forecasts. Complementing the predictive model is a chatbot, which serves as a user-friendly interface. This chatbot provides real-time stock prices for trending tickers, allows users to request on-demand predictions for specific stocks, and explains the complex model's architecture. It offers a comprehensive solution for financial analysis by combining advanced predictive capabilities with an intuitive, conversational user experience.

Key Words: Stock Price Prediction, BiLSTM, Improved Transformer, Temporal Convolutional Network (TCN), Deep Learning, Time-Series Forecasting, FinTech.

Guide **Project Coordinator** **HOD**

Guidelines for a Strong Title:

1. **Be Specific:** The title should clearly indicate the focus of the project. Avoid vague or overly broad terms.
2. **Include Key Elements:** Highlight core concepts
3. **Be Concise:** Aim for a title that is succinct yet descriptive. Typically, a title should be between 10-15 words.
4. **Use Keywords:** Include important keywords that reflect the core of your project. This helps in making the title more searchable and relevant.

Example Title Components:

1. **Technology or Approach:** Artificial Intelligence (AI) & Machine Learning (ML),Deep Learning
2. **Application Area:** Financial Technology (FinTech), Stock Market Forecasting, Time-Series Prediction.
3. **Purpose or Goal:** To build a hybrid deep learning model (BiLSTM + Improved Transformer) that improves accuracy, reduces errors, and provides robust stock price forecasts for better investment decisions.

Example Titles:

1. "Stock Price Prediction Using BiLSTM and Improved Transformer with Temporal Convolution"
2. "A Hybrid BiLSTM–Transformer Model for Robust Stock Market Forecasting" 3. "Deep Learning Framework for Financial Time-Series Prediction Using BiLSTM and Enhanced Transformer"
4. "BiLSTM and Transformer-TCN Hybrid Approach for Accurate Stock Price Forecasting".

Crafting a Title for the Provided Example:

StableStock - AI-Powered Stock Prediction & Trading Assistant

- **Technology Used:** BiLSTM and Improved Transformer
- **Main Focus:** Stock Price Prediction
- **Goal:** Improve accuracy and robustness in forecasting
- **Application Area:** Financial Markets / Stock Market Forecasting